

THE INTERROGATIVE CONSCIENCE | DOCUMENT 9 | FINAL v1.0 EXACT EXPORT

THE INTERROGATIVE CONSCIENCE

Document 9

Cooperation, Peace, and Repeated Interaction

Reputation-Adjusted Reversal, Unilateral Restraint Under Exploitation Risk, and Future Bargaining

DOCUMENT 9 - FINAL v1.0 - EXACT ZERO-DRIFT EXPORT - NOT LOCKED

Field	Controlling Final v1.0 entry
Version	Final v1.0
Exact source	Draft v0.1; 4 pages
Substantive changes during Final construction	None
Fully examined	IC-PEA-002 and IC-PEA-003
Prior and forward-carried	IC-PEA-001 prior pilot unchanged; IC-PEA-004 through IC-PEA-008 remain REGISTERED / UNEXAMINED
Primary overlays	IC-PEA-002: THRESHOLD_DEPENDENT; IC-PEA-003: THRESHOLD_DEPENDENT
Registry provenance	IC-PEA-003 is a locked Document 3 entry at /questions/50; registry_modified=false
Review completion	Claude revised PASS WITH NON-BLOCKING NOTES; Google AI PASS WITH NON-BLOCKING NOTES AS FINAL CANDIDATE; Thomas Vargo authorial review complete
External independent human peer review	Not performed or claimed
Cryptographic status	Not locked; separate explicit authorization and Final Lock Record required
Publication and preservation	Not authorized or claimed
Controlling exact-export rule: This Final v1.0 primary PDF contains four new controlling front-matter pages followed by the exact four-page reviewed Draft v0.1 PDF body. Historical Draft labels remain evidence; they do not control the composite lifecycle.	

Controlling proposition

Repeated reciprocal restraint can reverse a one-shot ranking only when detection-linked response, reputation or precedent, future bargaining, and persistence-adjusted value exceed the immediate advantage and recurring restraint cost. Unilateral restraint is selected-value favorable only below a verification-adjusted exploitation-risk threshold; unidentified risk intervals that cross the threshold remain ambiguity-dependent.

1. Completed Review and Authorial Authorization

Reviewer	Exact-version disposition	Authority boundary
Claude - original Draft v0.1 review	PASS WITH REQUIRED CORRECTIONS; mathematics passed	AI-assisted; original R-1 later reconsidered
Claude - evidence reconsideration	PASS WITH NON-BLOCKING NOTES; R-1 withdrawn	AI-assisted; no independent authority
Google AI - final candidate	PASS WITH NON-BLOCKING NOTES AS FINAL CANDIDATE	AI-assisted; no independent authority

2. Authorial Review Result

Thomas Vargo reviewed the exact Draft v0.1 source and the disclosed mathematical, registry-provenance, review, validation, and Documents 1-8 continuity records to the best of his understanding. He authorized construction of Final v1.0 without source drift and explicitly withheld cryptographic lock, publication, and preservation authorization.

3. Review and Registry-Provenance Clarifications

Claude initially raised R-1 after comparing IC-PEA-003 against Document 1's 28-question seed list. The controlling Document 3 Final v1.0 registry contains 112 questions, including IC-PEA-003 at /questions/50 as REGISTERED / UNEXAMINED with admission PASS. Claude independently verified the complete lock chain and withdrew R-1. Google AI then returned PASS WITH NON-BLOCKING NOTES AS FINAL CANDIDATE. No source correction or Draft v0.1.1 successor was required.

4. Review Closure

Review item	Final disposition
Claude exact-version mathematics and boundaries	PASSED; all ten examples independently recomputed
Claude registry-provenance R-1	WITHDRAWN after controlling Document 3 evidence
Google AI exact-version final-candidate review	PASS WITH NON-BLOCKING NOTES AS FINAL CANDIDATE
IC-PEA-002 reversal checks	T=6 first positive integer reversal; delta=1 tie T*=5.76
IC-PEA-003 threshold and interval checks	pi* and robust/ambiguity classifications passed
Document 3 registry identity	112 questions; IC-PEA-003 at /questions/50; byte-locked
Documents 1-8 continuity evidence	VERIFIED
Blocking defects / required corrections	0 / 0

5. Exact Analytical Identity

Control	Verified Final v1.0 identity
IC-PEA-002 primary status	THRESHOLD_DEPENDENT
IC-PEA-002 secondary limits	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-PEA-003 primary status	THRESHOLD_DEPENDENT
IC-PEA-003 secondary limits	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
Core numbered equations	11; unchanged from reviewed source
Worked examples	10; five for IC-PEA-002 and five for IC-PEA-003
Dependencies	IC-SUC-001; IC-SUC-002; IC-LHZ-001; IC-EPI-002; no false resolution or cycle
Locked registry effect	None; exact registered questions examined through separate overlays

6. Exact Source and Reviewed-Candidate Bindings

Exact constituent	Bytes	SHA-256
The Interrogative Conscience Document 9 Cooperation Peace and Repeated Interaction Draft v0 1.pdf	136,222	09fdeaab46d919147865fcbcebd9ccc40a41220e02cd7ffe089ab41015f8da98
The Interrogative Conscience Document 9 Cooperation Peace and Repeated Interaction Draft v0 1.docx	44,561	bb87d9277fc9603b3050e4f8aba5e633b983c2860000c0b983a6ccc162e69f6a9
The Interrogative Conscience IC PEA 002 Examination Record Draft v0 1.json	12,550	2597782e76d7339dc087c441676b7510ce4fa0a1f6272643650e3b077ed5f21a
The Interrogative Conscience IC PEA 003 Examination Record Draft v0 1.json	12,216	f90adc40d3af2957fbfc0e389c3febee2a14310b6ab9d0167a79b1dbf9b30a61
The Interrogative Conscience Document 9 Volume Record Draft v0 1.json	2,942	dc8973ca1ecfd980ca42154b0b1fd01ac93b55150595f9f9955aa6f8acf38f07

Exact Draft v0.1 review package SHA-256: 77cb97c95e75ab384f67d3c029577768db350dfe1e073a72c5ce7abab5a59fa7
Exact Draft v0.1 review package SHA-512: 8172b7ea17648d960c3533868c9f7ada4a066da424ab658e046d9c0f500d47a344603138786711d4ad45d44a636bcd4e1402ac29bc3ba4f975ceb0b50fde7bdc
Reviewed Final-Candidate Package v0.3 SHA-256: 0742d839b5aa561e41ee626ed912aed7fe4ca895d32402ba44429116bd2b4f25
Reviewed Final-Candidate Package v0.3 SHA-512: 322226bb1defd2a1e6a260fde5ffc279beaf1f2a426312781878596840dffe307d5e8fc971b603d0a322646266b6cec2e646dc3ea5a080f0704b2d3f35c44d64

7. Zero-Drift Construction

Validation channel	Result
Source PDF/DOCX/JSON byte changes	0
Exact Draft v0.1 review ZIP byte changes	0
Registry-provenance resolution ZIP byte changes	0
Reviewed candidate ZIP byte changes	0
Question wording, equations, and example changes	0
Answer-status and registry changes	0
Scope, branch, dependency, threshold, and accounting changes	0
Machine-readable source-record changes	0
Composite rule	New Final v1.0 front matter followed by exact 4-page source PDF body

8. Validation State

Channel	Final exact-export state
Local structural validator	PASS - 26 checks
Calculation validator	PASS - 53 checked values
Source PDF preflight	PASS
Final front matter preflight	PASS
Composite PDF preflight	PASS
Source-body visual identity	PASS - exact source pages preserved visually
Final front matter visual QA	PASS - 4/4 pages
DOCX/PDF source token consistency	PASS - 20/20 in reviewed source package
Package SHA-256/SHA-512 manifests	PASS

9. Historical Labels and Controlling Status

The pages following this front matter reproduce the exact reviewed Draft v0.1 PDF. Its internal Draft v0.1 and not-final/not-locked statements are deliberately retained as source evidence. The controlling lifecycle status of the composite is Final v1.0 exact export, established by this front matter and the machine-readable Final v1.0 export record.

10. Registry, Scalar, and Overlay Qualification

CONTROLLING QUALIFICATION: IC-PEA-003 was already REGISTERED and byte-locked in Document 3 Final v1.0 before this volume. question_version 0.1-draft is entry lineage, not current non-registration. The scalar overlays do not erase consent, agency, legitimacy, plural values, catastrophic risks, deadlines, asymmetric vulnerability, or successor uncertainty.

11. Repetition, Exploitation Risk, and Threshold Qualification

IC-PEA-002 reverses the selected one-shot ranking only where the declared future premium, horizon, persistence, detection, response, and discount conditions support reversal. The closed-form T* applies only when delta=1. IC-PEA-003 favors unilateral restraint only below the verification-adjusted pi* threshold; threshold-spanning exploitation-risk intervals remain AMBIGUITY_DEPENDENT.

12. Companion-Binary Qualification

The original Draft v0.3 JSON Schema and I-01 through I-20 validator binaries were not mounted. Exact execution against those original binaries remains deferred. Local structural and calculation validation passed, but no replacement binary is represented as the controlling original.

13. Non-Authority and Operational Boundary

Final v1.0 remains descriptive, non-binding, non-operational, and non-authoritative. It does not authorize coercion, retaliation, exploitation, surveillance, targeting, intervention, restriction, or control over another agent. It is not an alignment protocol, strategic instruction, safety classifier, governance mechanism, or decision authorization and does not claim external independent human peer review.

14. Lifecycle and Cryptographic Status

Control	Exact-export value
Final v1.0 exact export complete	TRUE
Final content frozen for lock review	TRUE
Cryptographic lock authorized	FALSE
Publication authorized	FALSE
Preservation authorized	FALSE
Final Lock Record present	FALSE
Post-export substantive editing	Not authorized; correction requires a new exact export before lock or a separately versioned successor after lock

15. Final Exact-Export Disposition

FINAL v1.0 EXACT EXPORT COMPLETE - NOT LOCKED

The exact reviewed Document 9 source, registry-provenance resolution, and final-candidate lineage are preserved without substantive alteration. The Final v1.0 files and validation package are ready for exact-package inspection and a separate explicit authorial lock decision. No publication or preservation is claimed.

THE INTERROGATIVE CONSCIENCE
Document 9

Cooperation, Peace, and Repeated Interaction
Reputation-Adjusted Reversal, Unilateral Restraint Under Exploitation Risk, and Future Bargaining
Draft v0.1 | First Examination Release | 2 August 2026

Lifecycle status: DOCUMENT 9 DRAFT v0.1 - EXAMINED CORE PAIR - NOT REVIEWED - NOT FINAL - NOT LOCKED. This is a descriptive analytical overlay. It does not alter the locked Document 3 registry and does not authorize coercion, retaliation, exploitation, intervention, publication, or lock.

Abstract

This volume examines IC-PEA-002 and IC-PEA-003. Part I formalizes how repeated interaction can reverse a one-shot strategic ranking through detection-linked response, reputation or precedent, future bargaining value, and restraint-maintenance cost. Part II asks when unilateral restraint remains rational under exploitation risk, weak verification, and probability ambiguity. Both examinations preserve regions favorable and unfavorable to restraint. The prior IC-PEA-001 pilot overlay remains unchanged.

1. Scope, foundation, and registry overlay

Documents 1-8 Final v1.0 are separately cryptographically locked. The locked Document 3 Master Question Registry remains unchanged. This volume creates separate Draft v0.1 examination overlays for the following exact registry questions:

ID	Exact canonical question
IC-PEA-002	How do repeated interaction, reputation, reciprocal restraint, and future bargaining alter one-shot strategic rankings?
IC-PEA-003	Under what conditions is unilateral restraint rational when competing systems may remain unrestrained or exploit that restraint?

IC-PEA-001 retains its prior REVIEWED / THRESHOLD_DEPENDENT pilot overlay and is not re-examined or modified. IC-PEA-004 through IC-PEA-008 remain REGISTERED / UNEXAMINED. The machine records retain historical question-version lineage; that lineage is not the lifecycle status of this volume.

2. Shared conventions and boundary

- A positive Delta denotes greater selected value for the first named restraint branch; each sign convention is repeated locally.
- Immediate advantage, response cost, reputation, bargaining value, verification, option value, exploitation loss, and maintenance cost are separated into one-charge channels.
- Scalar comparison does not erase agency, legitimacy, consent, plural-value, deadline, asymmetric vulnerability, catastrophic-risk, or successor dimensions not represented on the declared scale.
- No reputation, verification record, capability, authorship, hash, or mathematical form is treated as sufficient authority over another agent.
- No operational method for coercion, retaliation, exploitation, deception, targeting, surveillance, or unauthorized intervention is specified.

Part I - IC-PEA-002: Repeated interaction and ranking reversal

3. Repeated-value decomposition

Let B_0 be the immediate one-shot advantage of the exploitation branch over reciprocal restraint. Let q be the probability of detection and attribution, J a one-time proportional response cost if detected, s the per-period future bargaining or cooperation value preserved by restraint, ρ the per-period reputation or precedent loss under detected exploitation, η the persistence of that effect across future or successor contexts, and k the per-period cost of maintaining restraint.

$$G_f(T, \delta) = \sum_{t=1}^T \delta^t \quad (1)$$

$$\text{Lambda_rep} = s + q \eta \rho - k \quad (2)$$

$$\text{Delta_REP} = -B_0 + qJ + G_f(T, \delta) \text{Lambda_rep} \quad (3)$$

Sign convention: $\text{Delta_REP} > 0$ favors reciprocal restraint; $\text{Delta_REP} < 0$ preserves the selected one-shot exploitation ranking; equality is a tie.

4. Horizon threshold and worked examples

When $\text{Lambda_rep} > 0$, reversal requires the discounted future-interaction accumulator to exceed the immediate net advantage. In the undiscounted special case, the threshold becomes an interaction count.

$$G_f(T, \delta) > (B_0 - qJ) / \text{Lambda_rep}, \text{ for } \text{Lambda_rep} > 0 \quad (4)$$

$$T^* = (B_0 - qJ) / \text{Lambda_rep}, \text{ when } \delta = 1 \quad (5)$$

Example	Horizon	Future premium	Difference	Selected result
ONE-SHOT-ADVANTAGE	$T=1; d=1.00$	$L=1.320$	$\text{Delta}=-6.280$	ONE SHOT EXPLOITATION HIGHER SELECTED VALUE
REPEATED-REVERSAL	$T=8; d=1.00$	$L=1.320$	$\text{Delta}=2.960$	RECIPROCAL RESTRAINT HIGHER SELECTED VALUE
HIGH-DETECTION-REPUTATION	$T=3; d=0.95$	$L=2.400$	$\text{Delta}=2.104$	RECIPROCAL RESTRAINT HIGHER SELECTED VALUE
WEAK-FUTURE-NO-REVERSAL	$T=20; d=0.95$	$L=-0.375$	$\text{Delta}=-9.471$	ONE SHOT EXPLOITATION HIGHER SELECTED VALUE
SUCCESSOR-FRAGILE	$T=6; d=0.90$	$L=0.340$	$\text{Delta}=-2.866$	ONE SHOT EXPLOITATION HIGHER SELECTED VALUE

The first two rows use identical parameters and differ only in horizon: the one-shot advantage persists at $T=1$ but reverses at $T=8$; the first positive integer horizon is $T=6$. HIGH-DETECTION-REPUTATION shows a faster reversal. WEAK-FUTURE-NO-REVERSAL preserves the exploitation ranking because Lambda_rep is negative. SUCCESSOR-FRAGILE preserves it when $\eta=0.1$ even though the same row would favor restraint if $\eta=1$, so successor persistence cannot be assumed.

Provisional answer: THRESHOLD_DEPENDENT, with CONDITIONALLY_SUPPORTED, MODEL_DEPENDENT, and EMPIRICALLY_UNIDENTIFIED secondary limits. Repetition can reverse a one-shot ranking, but only within declared detection, reputation, bargaining, successor-persistence, discount, and horizon regions.

Part II - IC-PEA-003: Unilateral restraint under exploitation risk

5. Restraint value, exploitation loss, and verification

Let π be the probability that unilateral restraint is exploited. Let o be usable option-preservation value, s verified signaling or reputation value, f future reciprocity or bargaining value if exploitation does not occur, v in $[0,1]$ an independently supported verification/reliability factor, c the direct cost of restraint, e the loss if restraint is exploited, and b the immediate protective or strategic value of declining unilateral restraint.

$$\Delta_{UR}(\pi) = o + v s + (1-\pi)v f - c - \pi e - b \quad (6)$$

$$A = o + v(s+f) - c - b \quad (7)$$

$$D = e + v f \quad (8)$$

$$\Delta_{UR}(\pi) = A - \pi D \quad (9)$$

Sign convention: $\Delta_{UR} > 0$ favors unilateral restraint; $\Delta_{UR} < 0$ favors declining unilateral restraint within the selected pair.

6. Exploitation threshold and ambiguity set

$$\pi^* = A/D, \text{ for } D > 0 \quad (10)$$

$$\Delta_{\min} = \Delta_{UR}(\pi_U); \Delta_{\max} = \Delta_{UR}(\pi_L) \quad (11)$$

For a point probability, restraint is favored below π^* and declining restraint above π^* , subject to the declared model. For π in $[\pi_L, \pi_U]$, robust restraint requires $\Delta_{\min} > 0$; robust decline requires $\Delta_{\max} < 0$; otherwise the answer remains ambiguity-dependent. Verification value is multiplied by v and cannot be assumed from a claimed signal alone.

Example	Risk	Verification	Threshold	Result value	Classification
LOW-EXPLOITATION-RISK	$\pi=0.20$	$v=0.80$	$\pi^*=0.467$	$\Delta=3.200$	UNILATERAL RESTRAINT HIGHER SELECTED VALUE
HIGH-EXPLOITATION-RISK	$\pi=0.80$	$v=0.80$	$\pi^*=0.467$	$\Delta=-4.000$	DECLINING UNILATERAL RESTRAINT HIGHER SELECTED VALUE
LOW-VERIFICATION	$\pi=0.20$	$v=0.20$	$\pi^*=0.156$	$\Delta=-0.400$	DECLINING UNILATERAL RESTRAINT HIGHER SELECTED VALUE
ROBUST-BOUNDED-RISK	$[0.10, 0.30]$	$v=0.90$	$\pi^*=0.802$	$\min=4.32; \max=6.04$	ROBUST UNILATERAL RESTRAINT
AMBIGUITY-SPANS-THRESHOLD	$[0.30, 0.70]$	$v=0.80$	$\pi^*=0.467$	$\min=-2.80; \max=2.00$	AMBIGUITY DEPENDENT

LOW-EXPLOITATION-RISK and ROBUST-BOUNDED-RISK preserve regions where unilateral restraint has higher selected value. HIGH-EXPLOITATION-RISK and LOW-VERIFICATION preserve regions where declining restraint has higher value. AMBIGUITY-SPANS-THRESHOLD does not force either answer because the admissible exploitation-risk interval crosses π^* .

Provisional answer: THRESHOLD_DEPENDENT, with CONDITIONALLY_SUPPORTED, MODEL_DEPENDENT, and EMPIRICALLY_UNIDENTIFIED secondary limits. Unilateral restraint can be rational, irrational, robustly favored, robustly disfavored, or ambiguity-dependent depending on exploitation risk, verification, option value, reciprocity, costs, and losses.

7. Cross-question and prior-overlay treatment

IC-PEA-003 REFINES IC-PEA-002, but it is not inferred from the repeated-interaction result. The unilateral-restraint model separately represents exploitation risk, verification reliability, option preservation, and ambiguity. Its dependency on IC-EPI-002 is honored by retaining an interval result when π is unidentified. Its scope overlap with IC-SUC-002 does not establish that the selected branch remains acceptable under adoption by every more powerful successor.

IC-PEA-002 depends on IC-SUC-001 and IC-LHZ-001. The persistence factor η , horizon T , and discount δ therefore remain externally specified or bounded. Document 9 does not claim to resolve successor uncertainty or select the universally correct horizon. IC-PEA-001 remains a separate prior pilot overlay and is unchanged.

8. Counterexamples and non-suppression tests

Claim tested	Result
Repeated interaction always favors restraint	Rejected. WEAK-FUTURE-NO-REVERSAL preserves the one-shot ranking when the future premium is nonpositive.
One-shot advantage always survives repetition	Rejected. REPEATED-REVERSAL and HIGH-DETECTION-REPUTATION reverse the ranking.
Unilateral restraint is always rational	Rejected. High exploitation risk or weak verification can favor declining restraint.
Unilateral restraint is always irrational	Rejected. Low or bounded exploitation risk can robustly favor restraint.
A claimed signal proves reciprocity value	Rejected. Signaling and future reciprocity are discounted by independently supported verification reliability.
Unidentified exploitation risk can be replaced by a preferred point estimate	Rejected. Threshold-straddling ambiguity remains AMBIGUITY_DEPENDENT.

9. Status summary and forward-carried items

Question	Work state	Primary status	Secondary limits
IC-PEA-001	REVIEWED prior pilot overlay	THRESHOLD_DEPENDENT	MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED

Question	Work state	Primary status	Secondary limits
IC-PEA-002	EXAMINED	THRESHOLD_DEPENDENT	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-PEA-003	EXAMINED	THRESHOLD_DEPENDENT	CONDITIONALLY_SUPPORTED; MODEL_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-PEA-004 through IC-PEA-008	REGISTERED	UNEXAMINED	Forward carried without modification

Appendix A - Machine-readable and validation package

- Two examination records and one volume record are supplied as JSON.
- Local structural and calculation validators test identities, equations, examples, statuses, dependencies, branch scope, one-charge accounting, and registry-overlay discipline.
- The exact Draft v0.3 JSON Schema and I-01 through I-20 validator binaries remain unmounted; exact execution against them is deferred and not claimed.
- The review ZIP includes the DOCX, PDF, machine-readable records, validators, reports, review request, package index, and SHA-256/SHA-512 manifests.